



Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI

Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Values

Review Rating column

\$59.76

Avg Purchase

Demographics · Purchase details · Shopping behavior · Subscription status

Data Preparation in Python

01

Load & Explore

pandas, df.info(), describe()

02

Clean

Impute missing ratings by category median

03

Engineer

age_group bins, purchase_frequency_days

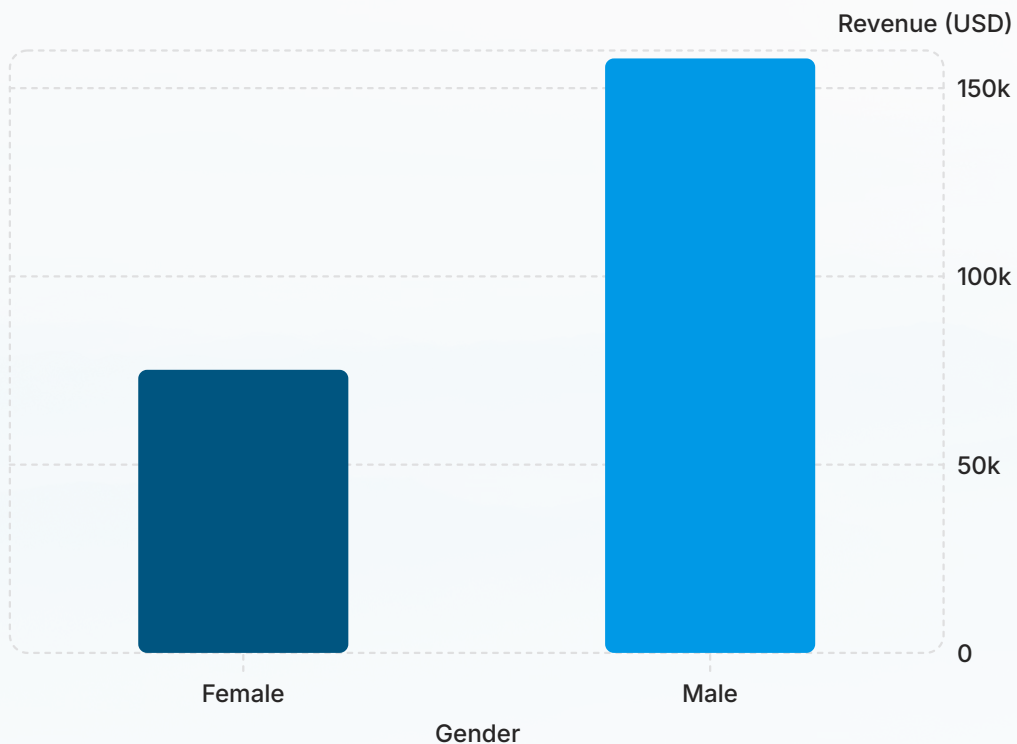
04

Load to DB

PostgreSQL for SQL analysis



Revenue by Gender



Key Finding

Male customers generated **2.1× more revenue** than female customers — a critical signal for targeting and acquisition strategy.

Product Insights

Top-Rated Products

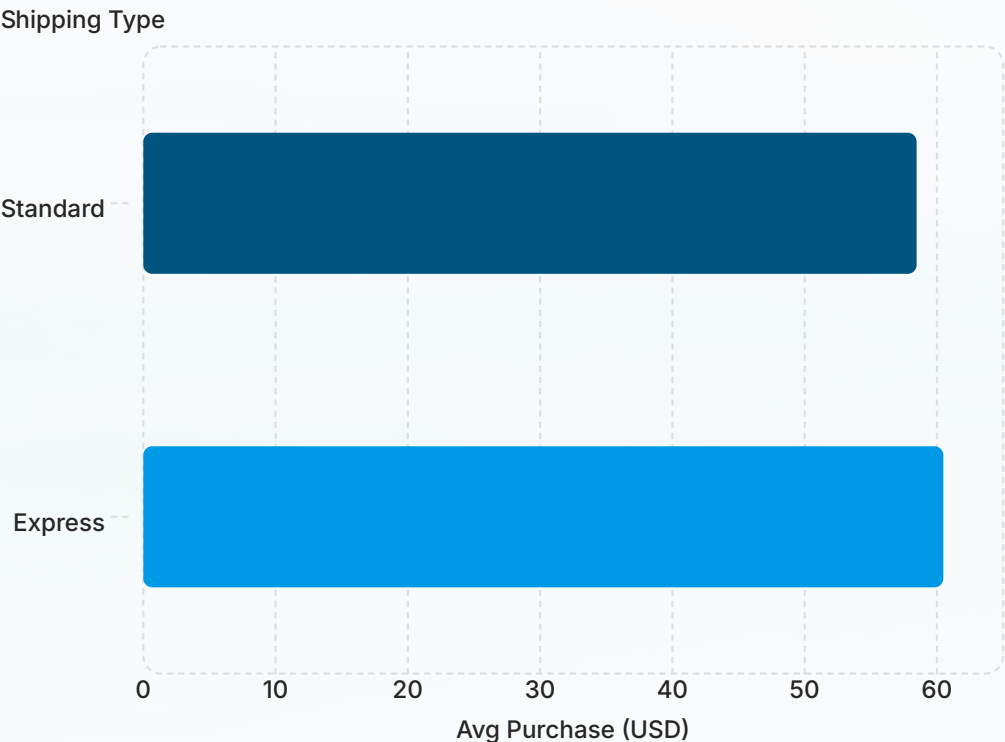
Item	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent

Item	Discount %
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Shipping & Subscriptions

Avg Purchase by Shipping Type

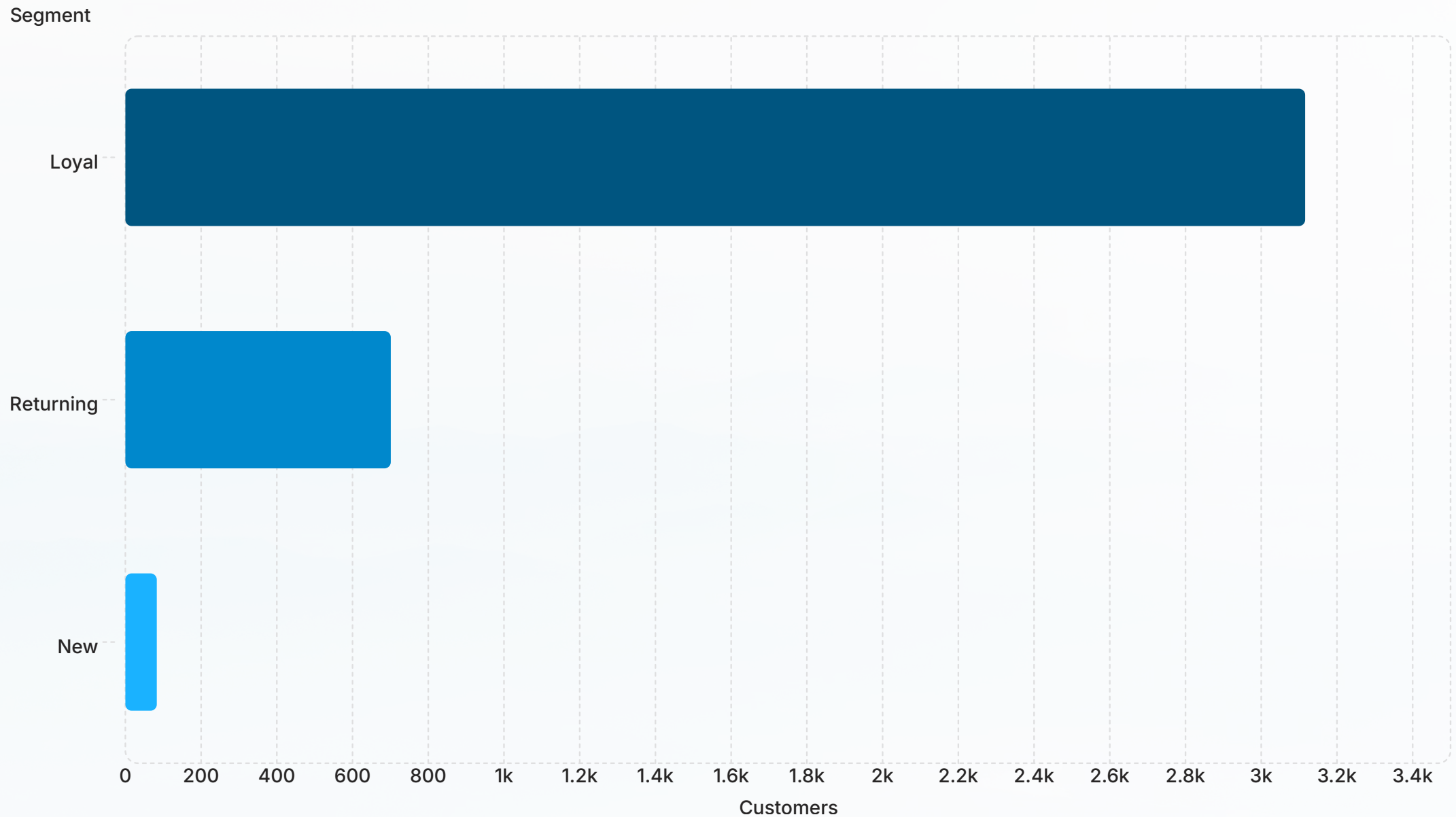


Subscribers vs Non-Subscribers

Status	Customers	Revenue
Yes	1,053	\$62,645
No	2,847	\$170,436

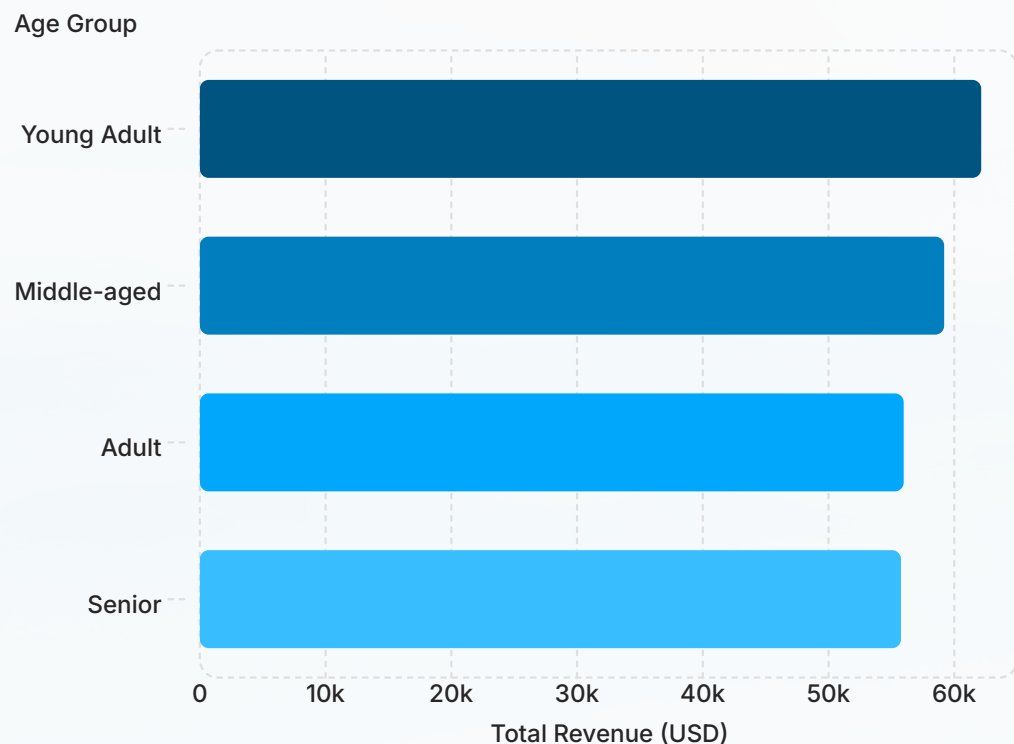
Non-subscribers drive **73% of revenue** — a major growth opportunity.

Customer Segmentation



80% of customers are **Loyal** (3,116), with 701 Returning and only 83 New customers. Repeat buyers: 958 subscribed vs 2,518 not subscribed.

Revenue by Age Group



Insight

Young Adults lead revenue contribution. Revenue is relatively balanced across all age groups — broad targeting is viable.



Power BI Dashboard

KPIs

3.9K customers · \$59.76 avg purchase · 3.75 avg rating

Filters

Subscription · Gender · Category · Shipping Type

Visuals

Revenue & Sales by Category and Age Group

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers → Loyal segment



Review Discounts

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express shippers